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Calibrating Sequence Likelihood Improves Conditional Language Generation

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Conditional Language Generation

Auto Regressive modeling

Input context $\mathbf{x}$ 가 주어졌을 때, target $\mathbf{y}$ 의 확률을 학습함으로써 모델링

$$P_{\theta}(\mathbf{y}|\mathbf{x}) = \prod_{t=0}^l P_{\theta}(y^t|y^0 \dots y^{t-1}, \mathbf{x})$$

$$L = \sum_i^N -\log(P_{\theta}(\mathbf{y}_i|\mathbf{x}_i)) = \sum_i^N \sum_{t=0}^l -\log(P_{\theta}(y_i^t|y_i^0 \dots y_i^{t-1}, \mathbf{x}_i))$$

Objective of Maximum Likelihood Estimation

배경 MLE Framework의 한계

Uncalibrated Sequence Likelihood

Text generation task의 데이터셋은 주로 **1개의 target sequence**로 구성

- 시퀀스들을 비교하는 직접적인 supervision이 부족하며, 모델의 generalization 능력에만 의존하게 됨
- MLE로 학습된 모델에 대해, probability와 시퀀스 품질 간 상관관계가 낮음
- Teacher forcing으로 인한 Exposure bias는 생성된 시퀀스의 품질을 더욱 악화시킬 수 있음

⇒ 시퀀스들 간의 비교를 통해 적절한 likelihood를 취할 능력이 부족함

Uncalibrated Sequence Likelihood

문제를 해결하기 위한 접근 방식

(1) Reinforcement Learning with sequence-level rewards

RL finetuning stage에서 ROUGE score, human judgement에 대해 직접적으로 최적화

(2) Two-stage system (Generator and Reranker)

Candidate들을 생성하는 generation 모델과, 이를 재정렬하는 reranking 모델이 별도 존재

(3) Multi-task learning with sequence-level loss

본 연구에서 채택한 접근법

Contrastive loss와 Cross entropy를 결합한 objective로 학습 (multi-task loss)

Q & A

- MLE Framework의 한계
- 한계를 극복하기 위한 접근법

SLiC (Sequence Likelihood Calibration)

Calibration stage를 도입함으로써, MLE framework로 학습된 모델의 calibration 능력을 보완

- Offline learning
- Evaluation metric을 학습 과정에 사용하지 않음
- 새로운 모델이 아닌, Fine-tuned model을 새로운 objective에 이어서 학습

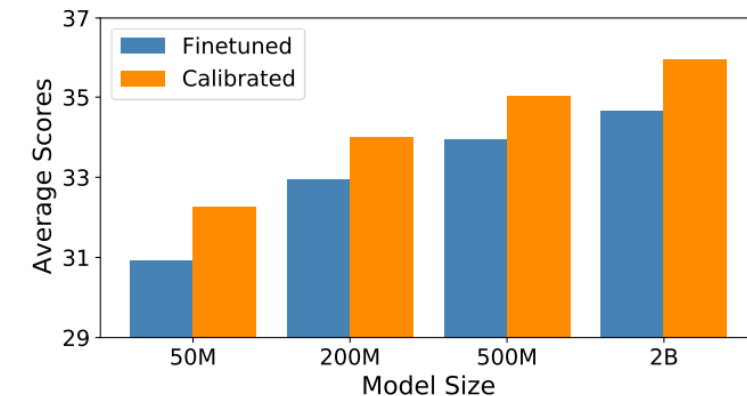
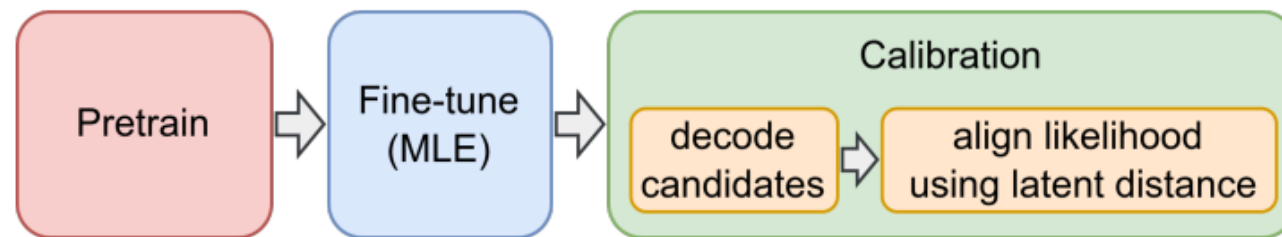


Figure 1. Calibrating sequence likelihood improves language generation across model scales.

SLiC (Sequence Likelihood Calibration)

Calibration stage

동작 알고리즘

Algorithm 1 Calibrating Sequence Likelihood

for $\mathbf{x}, \bar{\mathbf{y}} \in \{\mathbf{x}, \bar{\mathbf{y}}\}_n$ **do**
 $\{\hat{\mathbf{y}} \sim P_{\theta_{ft}}(\mathbf{y}|\mathbf{x})\}_m$

Fine-tuned model로부터 m개의 candidate decoding

$\theta \leftarrow \theta_{ft}$

for $\{\mathbf{x}, \bar{\mathbf{y}}, \{\hat{\mathbf{y}}\}_m\}_b \sim \{\mathbf{x}, \bar{\mathbf{y}}, \{\hat{\mathbf{y}}\}_m\}_n$ **do**
 $\theta \leftarrow \theta - lr \nabla_{\theta} \mathcal{L}(\theta)$

SLiC (Sequence Likelihood Calibration)

Calibration stage

동작 알고리즘

Algorithm 1 Calibrating Sequence Likelihood

for $\mathbf{x}, \bar{\mathbf{y}} \in \{\mathbf{x}, \bar{\mathbf{y}}\}_n$ **do**
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$\theta \leftarrow \theta_{ft}$
 for $\{\mathbf{x}, \bar{\mathbf{y}}, \{\hat{\mathbf{y}}\}_m\}_b \sim \{\mathbf{x}, \bar{\mathbf{y}}, \{\hat{\mathbf{y}}\}_m\}_n$ **do**
 $\theta \leftarrow \theta - lr \nabla_{\theta} \mathcal{L}(\theta)$

Fine-tuned model을 새로운 objective $\mathcal{L}(\theta)$ 에 학습

$$\mathcal{L}(\theta) = \sum_b L^{\text{cal}}(\theta, s; \mathbf{x}, \bar{\mathbf{y}}, \{\hat{\mathbf{y}}\}_m) + \lambda L^{\text{reg}}(\theta, \theta_{ft}; \mathbf{x}, \bar{\mathbf{y}})$$

SLiC (Sequence Likelihood Calibration)

Calibration stage : Objective

$$\mathcal{L}(\theta) = \sum_b \underbrace{L^{\text{cal}}(\theta, s; \mathbf{x}, \bar{\mathbf{y}}, \{\hat{\mathbf{y}}\}_m)}_{\text{Calibration loss}} + \lambda \underbrace{L^{\text{reg}}(\theta, \theta_{ft}; \mathbf{x}, \bar{\mathbf{y}})}_{\text{Regularization loss}}$$

Calibration loss

모델의 candidate sequence likelihood를 target sequence와의 유사도에 따라 정렬하기 위함

Regularization loss

기존의 MLE objective에서 크게 벗어나지 않도록 규제하기 위함

S Similarity function

$\mathbf{x}$ Input context

$\bar{\mathbf{y}}$ Target sequence

$\{\hat{\mathbf{y}}\}_m$ m 개의 candidates

SLiC (Sequence Likelihood Calibration)

Calibration stage : Calibration loss

Similarity function s

Zhang et al*의 연구에서 제안된 수식을 활용했으며,

$n = 1, 2, 4, 8$ 에 대해 candidate과 target 간 코사인 유사도로부터 F1 score 계산한 값

$$s_{\theta}(\hat{\mathbf{y}}, \bar{\mathbf{y}}; \mathbf{x}) = \sum_n F_n(\hat{\mathbf{e}}, \bar{\mathbf{e}})$$

$$\hat{\mathbf{e}} = emb(\hat{\mathbf{y}}, \mathbf{x})$$

$$\bar{\mathbf{e}} = emb(\bar{\mathbf{y}}, \mathbf{x})$$

$emb(\mathbf{y}, \mathbf{x})$:

$\mathbf{x}$ 와 $\mathbf{y}$ 가 주어졌을 때 Decoder hidden state

* Zhang, T., et al. "Bertscore: Evaluating text generation with bert." *Proceeding of the 8th International Conference on Learning Representations (ICLR 2020)*, Addis Ababa, Ethiopia, 2020.

SLiC (Sequence Likelihood Calibration)

Calibration stage : Calibration loss

Similarity function S

Zhang et al의 연구에서 제안된 수식을 활용했으며, 두 벡터의 코사인 유사도로부터 F1 score 계산

F1 score

$$F_n = 2 \frac{P_n \times R_n}{P_n + R_n}$$

Precision

$$P_n(\hat{\mathbf{e}}, \bar{\mathbf{e}}) = \frac{1}{|\hat{\mathbf{e}}|} \sum_{\hat{\mathbf{e}}_{i:i+n}} \max_{\bar{\mathbf{e}}_{j:j+n}} \hat{\mathbf{e}}_{i:i+n}^T \bar{\mathbf{e}}_{j:j+n}$$

Recall

$$R_n(\hat{\mathbf{e}}, \bar{\mathbf{e}}) = \frac{1}{|\bar{\mathbf{e}}|} \sum_{\bar{\mathbf{e}}_{j:j+n}} \max_{\hat{\mathbf{e}}_{i:i+n}} \hat{\mathbf{e}}_{i:i+n}^T \bar{\mathbf{e}}_{j:j+n}$$

SLiC (Sequence Likelihood Calibration)

Calibration stage : Calibration loss

Similarity function S

Precision

$$P_n(\hat{\mathbf{e}}, \bar{\mathbf{e}}) = \frac{1}{|\hat{\mathbf{e}}|} \sum_{\hat{\mathbf{e}}_{i:i+n}} \max_{\bar{\mathbf{e}}_{j:j+n}} \hat{\mathbf{e}}_{i:i+n}^T \bar{\mathbf{e}}_{j:j+n}$$

Candidate에 나타난 n개의 token에 대해,
Target의 모든 n개의 token들과 유사도 계산

Recall

$$R_n(\hat{\mathbf{e}}, \bar{\mathbf{e}}) = \frac{1}{|\bar{\mathbf{e}}|} \sum_{\bar{\mathbf{e}}_{j:j+n}} \max_{\hat{\mathbf{e}}_{i:i+n}} \hat{\mathbf{e}}_{i:i+n}^T \bar{\mathbf{e}}_{j:j+n}$$

Target에 나타난 n개의 token에 대해,
Candidate의 모든 n개의 token들과 유사도 계산

SLiC (Sequence Likelihood Calibration)

Calibration stage : Calibration loss

4가지 유형의 loss 사용

Positive candidate과 Negative candidate 간 차이를 극대화하는 방향으로 학습

$$\text{Rank loss} \quad \max(0, \beta - \log P_{\theta}(\hat{\mathbf{y}}_+ | \mathbf{x}) + \log P_{\theta}(\hat{\mathbf{y}}_- | \mathbf{x}))$$

$$\text{Margin loss} \quad \max(0, \beta(s(\hat{\mathbf{y}}_+, \bar{\mathbf{y}}; \mathbf{x}) - s(\hat{\mathbf{y}}_-, \bar{\mathbf{y}}; \mathbf{x})) - \log P_{\theta}(\hat{\mathbf{y}}_+ | \mathbf{x}) + \log P_{\theta}(\hat{\mathbf{y}}_- | \mathbf{x}))$$

$$\text{List-wise rank loss} \quad \sum_{i < j} \max(0, \beta|i - j| - \log P_{\theta}(\hat{\mathbf{y}}_i | \mathbf{x}) + \log P_{\theta}(\hat{\mathbf{y}}_j | \mathbf{x}))$$

$$\text{Expected reward loss} \quad \sum_i \left[-s(\hat{\mathbf{y}}_i, \bar{\mathbf{y}}; \mathbf{x}) * \frac{P_{\theta}(\hat{\mathbf{y}}_i | \mathbf{x})}{\sum_i P_{\theta}(\hat{\mathbf{y}}_i | \mathbf{x})} \right]$$

SLiC (Sequence Likelihood Calibration)

Calibration stage : Calibration loss

Rank loss $\max(0, \beta - \log P_{\theta}(\hat{\mathbf{y}}_+ | \mathbf{x}) + \log P_{\theta}(\hat{\mathbf{y}}_- | \mathbf{x}))$ Positive candidate가 negative보다
높은 확률을 갖도록 학습

Margin loss $\max(0, \beta(s(\hat{\mathbf{y}}_+, \bar{\mathbf{y}}; \mathbf{x}) - s(\hat{\mathbf{y}}_-, \bar{\mathbf{y}}; \mathbf{x})) - \log P_{\theta}(\hat{\mathbf{y}}_+ | \mathbf{x}) + \log P_{\theta}(\hat{\mathbf{y}}_- | \mathbf{x}))$

List-wise rank loss $\sum_{i < j} \max(0, \beta|i - j| - \log P_{\theta}(\hat{\mathbf{y}}_i | \mathbf{x}) + \log P_{\theta}(\hat{\mathbf{y}}_j | \mathbf{x}))$

Expected reward loss $\sum_i \left[-s(\hat{\mathbf{y}}_i, \bar{\mathbf{y}}; \mathbf{x}) * \frac{P_{\theta}(\hat{\mathbf{y}}_i | \mathbf{x})}{\sum_i P_{\theta}(\hat{\mathbf{y}}_i | \mathbf{x})} \right]$

SLiC (Sequence Likelihood Calibration)

Calibration stage : Calibration loss

Similarity 차이만큼

Positive candidate가 negative보다

높은 확률을 갖도록 학습

Rank loss $\max(0, \beta - \log P_{\theta}(\hat{y}_+ | \mathbf{x}) + \log P_{\theta}(\hat{y}_- | \mathbf{x}))$

Margin loss $\max(0, \beta(s(\hat{y}_+, \bar{y}; \mathbf{x}) - s(\hat{y}_-, \bar{y}; \mathbf{x}))) - \log P_{\theta}(\hat{y}_+ | \mathbf{x}) + \log P_{\theta}(\hat{y}_- | \mathbf{x}))$

List-wise rank loss $\sum_{i < j} \max(0, \beta|i - j| - \log P_{\theta}(\hat{y}_i | \mathbf{x}) + \log P_{\theta}(\hat{y}_j | \mathbf{x}))$

Expected reward loss $\sum_i \left[-s(\hat{y}_i, \bar{y}; \mathbf{x}) * \frac{P_{\theta}(\hat{y}_i | \mathbf{x})}{\sum_i P_{\theta}(\hat{y}_i | \mathbf{x})} \right]$

SLiC (Sequence Likelihood Calibration)

Calibration stage : Calibration loss

Rank loss $\max(0, \beta - \log P_{\theta}(\hat{\mathbf{y}}_+ | \mathbf{x}) + \log P_{\theta}(\hat{\mathbf{y}}_- | \mathbf{x}))$

Margin loss $\max(0, \beta(s(\hat{\mathbf{y}}_+, \bar{\mathbf{y}}; \mathbf{x}) - s(\hat{\mathbf{y}}_-, \bar{\mathbf{y}}; \mathbf{x})) - \log P_{\theta}(\hat{\mathbf{y}}_+ | \mathbf{x}) + \log P_{\theta}(\hat{\mathbf{y}}_- | \mathbf{x}))$

List-wise rank loss $\sum_{i < j} \max(0, \beta|i - j| - \log P_{\theta}(\hat{\mathbf{y}}_i | \mathbf{x}) + \log P_{\theta}(\hat{\mathbf{y}}_j | \mathbf{x}))$

Similarity rank 차이만큼

Positive candidate이 negative보다

높은 확률을 갖도록 학습

Expected reward loss $\sum_i \left[-s(\hat{\mathbf{y}}_i, \bar{\mathbf{y}}; \mathbf{x}) * \frac{P_{\theta}(\hat{\mathbf{y}}_i | \mathbf{x})}{\sum_i P_{\theta}(\hat{\mathbf{y}}_i | \mathbf{x})} \right]$

SLiC (Sequence Likelihood Calibration)

Calibration stage : Calibration loss

Rank loss $\max(0, \beta - \log P_\theta(\hat{\mathbf{y}}_+ | \mathbf{x}) + \log P_\theta(\hat{\mathbf{y}}_- | \mathbf{x}))$

Margin loss $\max(0, \beta(s(\hat{\mathbf{y}}_+, \bar{\mathbf{y}}; \mathbf{x}) - s(\hat{\mathbf{y}}_-, \bar{\mathbf{y}}; \mathbf{x})) - \log P_\theta(\hat{\mathbf{y}}_+ | \mathbf{x}) + \log P_\theta(\hat{\mathbf{y}}_- | \mathbf{x}))$

List-wise rank loss $\sum_{i < j} \max(0, \beta|i - j| - \log P_\theta(\hat{\mathbf{y}}_i | \mathbf{x}) + \log P_\theta(\hat{\mathbf{y}}_j | \mathbf{x}))$

Expected reward loss $\sum_i \left[-s(\hat{\mathbf{y}}_i, \bar{\mathbf{y}}; \mathbf{x}) * \frac{P_\theta(\hat{\mathbf{y}}_i | \mathbf{x})}{\sum_i P_\theta(\hat{\mathbf{y}}_i | \mathbf{x})} \right]$ Similarity가 가장 높은 candidate이
높은 확률을 가지도록 학습

SLiC (Sequence Likelihood Calibration)

Calibration stage : Regularization loss

2가지 loss를 사용했으며, 각 loss에 따른 비교 실험 진행

Cross entropy $\sum_t -\log P_{\theta}(\bar{y}_t | \bar{\mathbf{y}}_{t-1}, \mathbf{x})$

KL divergence $\sum_t P_{\theta}(\bar{y}_t | \bar{\mathbf{y}}_{t-1}, \mathbf{x}) \log \frac{P_{\theta}(\bar{y}_t | \bar{\mathbf{y}}_{t-1}, \mathbf{x})}{P_{\theta_{ft}}(\bar{y}_t | \bar{\mathbf{y}}_{t-1}, \mathbf{x})}$

Fine-tuned model과 calibrated model의
확률 분포의 차이를 낮춤

2022

Calibrating Sequence Likelihood Improves Conditional Language Generation

Q & A

- Sequence Likelihood Calibration (SLiC)

Experiments

실험 구성

1. Ablation Studies of Calibration

Calibration stage를 추가했을 때, 성능 개선이 이루어졌는가?

2. Benefits of Calibrated sequence likelihood

Fine-tuned only model과 비교했을 때, SLiC의 장점은 무엇인가?

3. Scaling Properties of Calibrated models

모델 크기 별 inference compute에 따른 generation 성능 비교

Experiments

Ablation Studies of Calibration

사전 학습된 PEGASUS_{2B} 모델에 대해 AB study 수행
Text summarization task에서
fine-tuned model과 calibrated model의 성능 비교

Δ : fine-tuned model 대비
calibrated model의 R_m 개선율

$$R_m = \frac{1}{4} \sum_d \sqrt[3]{R_1 R_2 R_L}$$

Ablation	CNN/DailyMail R1 / R2 / RL	XSUM R1 / R2 / RL	RedditTIFU-long R1 / R2 / RL	SAMSum R1 / R2 / RL	Δ avg
fine-tuned	44.74/21.83/41.92	47.23/24.31/39.12	26.84/9.08/21.92	53.67/29.35/44.75	0.00%
<i>similarity function</i>					
ROUGE	46.47/22.49/43.63	47.86/24.55/39.58	29.92/9.83/23.93	54.82/30.15/45.30	3.26%
decoder repr	46.55/22.50/43.69	47.88/24.62/39.62	29.86/9.84/23.91	54.72/29.96/45.10	3.20%
token emb	46.51/22.48/43.67	47.04/23.63/38.39	29.78/9.69/23.46	53.71/29.38/44.75	1.64%
<i>calibration loss</i>					
rank	46.73/22.70/43.85	48.11/24.80/40.06	30.34/9.80/24.32	55.19/30.46/46.32	4.27%
margin	46.11/22.46/43.30	47.62/24.81/39.89	30.84/9.97/24.37	54.58/30.10/45.92	3.63%
list rank	46.62/22.88/43.76	47.93/24.57/39.67	30.87/9.65/24.46	54.56/29.81/45.17	3.49%
reward	46.49/22.55/43.63	47.77/24.48/39.49	30.99/9.95/24.39	54.42/29.98/45.56	3.47%
<i>regularization loss</i>					
none	46.54/22.44/43.68	47.51/24.70/39.82	30.73/9.68/24.05	55.07/30.07/45.60	3.48%
cross entropy	46.73/22.70/43.85	48.11/24.80/40.06	29.96/9.72/23.82	55.19/30.46/46.32	4.06%
KL divergence	46.80/22.83/43.98	47.96/24.92/40.09	30.73/9.68/24.05	54.87/30.20/45.95	4.09%
<i>candidates decoding method</i>					
beam search	46.50/22.48/43.66	47.82/24.65/39.67	31.04/9.96/24.37	54.66/30.27/45.46	3.70%
diverse beam	46.31/22.48/43.47	47.79/24.53/39.51	31.00/9.95/24.08	54.57/29.67/45.55	3.26%
nucleus	46.45/22.46/43.54	47.67/24.50/39.47	31.09/10.01/24.31	54.61/30.04/45.63	3.51%
<i>calibration checkpoint selection</i>					
ROUGE	46.66/22.66/43.84	48.03/24.78/39.79	30.94/9.98/24.43	54.63/30.03/45.79	3.96%
perplexity	47.36/24.02/44.45	47.96/24.74/39.78	31.04/10.08/24.53	54.65/30.11/46.00	4.93%

Table 1. Ablation of the sequence likelihood calibration method.

Experiments

Ablation Studies of Calibration

(1) Similarity function

ROUGE

ROUGE score에 기반하여, $\hat{y}$ 와 y 의 유사도 계산
Evaluation metric에 직접적으로 최적화하는 위험이 있다.

Decoder hidden state

본 연구에서 제안한 방식으로,
input context x 와 target sequence y 에 대한 임베딩을 얻을 수 있다

Token embedding of y

y 와 $\hat{y}$ 의 token embedding으로부터 코사인 유사도를 계산하여
F1 score를 도출

Ablation	CNN/DailyMail R1 / R2 / RL	XSUM R1 / R2 / RL	RedditTIFU-long R1 / R2 / RL	SAMSum R1 / R2 / RL	Δ avg
fine-tuned	44.74/21.83/41.92	47.23/24.31/39.12	26.84/9.08/21.92	53.67/29.35/44.75	0.00%
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token emb	46.51/22.48/43.67	47.04/23.63/38.39	29.78/9.69/23.46	53.71/29.38/44.75	1.64%

Table 1. Ablation of the sequence likelihood calibration method.

Calibration loss : Expected reward
Regularization loss : cross entropy
Decoding method : beam search (size=15)
Checkpoint selection: ROUGE

Experiments

Ablation Studies of Calibration

(2) Calibration loss

Calibration loss 유형에 따른 성능 비교

Rank loss를 사용했을 때, evaluation 성능이 가장 높음

Regularization loss : cross entropy

Decoding method : beam search (size=15)

Checkpoint selection: ROUGE

(3) Regularization loss

Regularization loss 유형에 따른 성능 비교

KL divergence를 사용했을 때, evaluation 성능이 가장 높았으며,

Cross entropy와 큰 차이를 보이지 않음

Calibration loss : rank loss

Decoding method : beam search (size=15)

Checkpoint selection: ROUGE

Ablation	CNN/DailyMail R1 / R2 / RL	XSUM R1 / R2 / RL	RedditTIFU-long R1 / R2 / RL	SAMSum R1 / R2 / RL	Δ avg
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Table 1. Ablation of the sequence likelihood calibration method.

Experiments

Ablation Studies of Calibration

(4) Candidates decoding method

m 개의 Candidate 디코딩 시, method에 따른 성능 비교
Beam search를 사용했을 때 가장 높은 성능을 보임

Calibration loss : Expected reward
Regularization loss : cross entropy
Checkpoint selection: ROUGE

(5) Calibration Checkpoint selection

Calibration stage에서 학습한 모델의 checkpoint를
선정하는 기준에 따른 성능 비교
Perplexity를 기준으로 했을 때 가장 높은 성능을 보임

Calibration loss : Expected reward
Regularization loss : cross entropy
Decoding method : beam search (size=15)

Ablation	CNN/DailyMail R1 / R2 / RL	XSUM R1 / R2 / RL	RedditTIFU-long R1 / R2 / RL	SAMSum R1 / R2 / RL	Δ avg
fine-tuned	44.74/21.83/41.92	47.23/24.31/39.12	26.84/9.08/21.92	53.67/29.35/44.75	0.00%
<i>candidates decoding method</i>					
beam search	46.50/22.48/43.66	47.82/24.65/39.67	31.04/9.96/24.37	54.66/30.27/45.46	3.70%
diverse beam	46.31/22.48/43.47	47.79/24.53/39.51	31.00/9.95/24.08	54.57/29.67/45.55	3.26%
nucleus	46.45/22.46/43.54	47.67/24.50/39.47	31.09/10.01/24.31	54.61/30.04/45.63	3.51%
<i>calibration checkpoint selection</i>					
ROUGE	46.66/22.66/43.84	48.03/24.78/39.79	30.94/9.98/24.43	54.63/30.03/45.79	3.96%
perplexity	47.36/24.02/44.45	47.96/24.74/39.78	31.04/10.08/24.53	54.65/30.11/46.00	4.93%

Table 1. Ablation of the sequence likelihood calibration method.

Experiments

Benefits of Calibrated sequence likelihood

Decoding method에 따른 Fine-tuned only model과 calibrated model의 성능 비교

Fine-tuned model은 candidate 수가 많아질수록 성능이 하락하는 반면, calibrated model은 향상

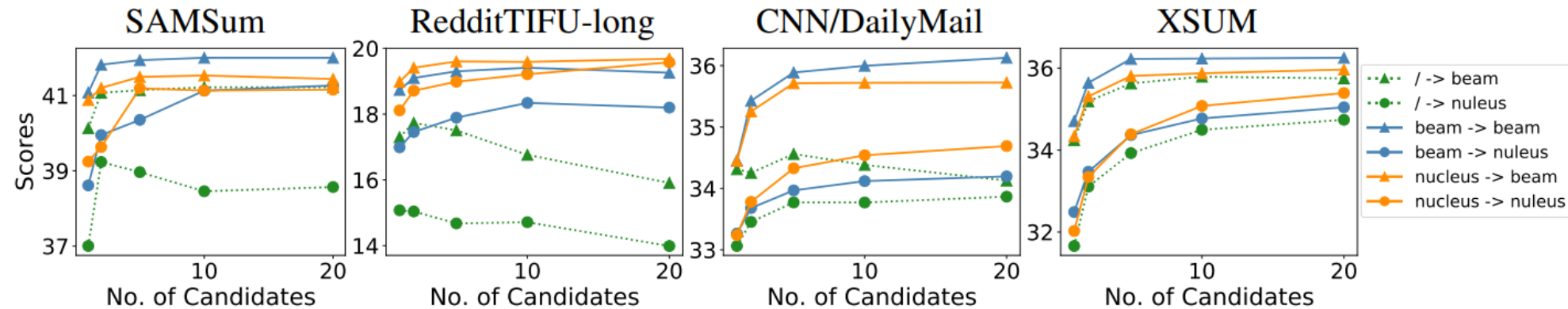


Figure 2. Effect of decoding methods on calibrated and fine-tuned only models.

Experiments

Scaling Properties of Calibrated models

모델 크기 별 Inference compute와 그에 따른 generation 성능 비교

- ⇒ Fine-tuned only model과 달리 Calibrated model은 candidate 수가 커질 수록 성능 향상
- ⇒ FLOPs가 커질수록 성능이 향상되는 경향이 있음
- ⇒ SAMSum과 CNN/DM 데이터셋에서는, 작은 모델이 큰 모델에 준하는 성능을 보임

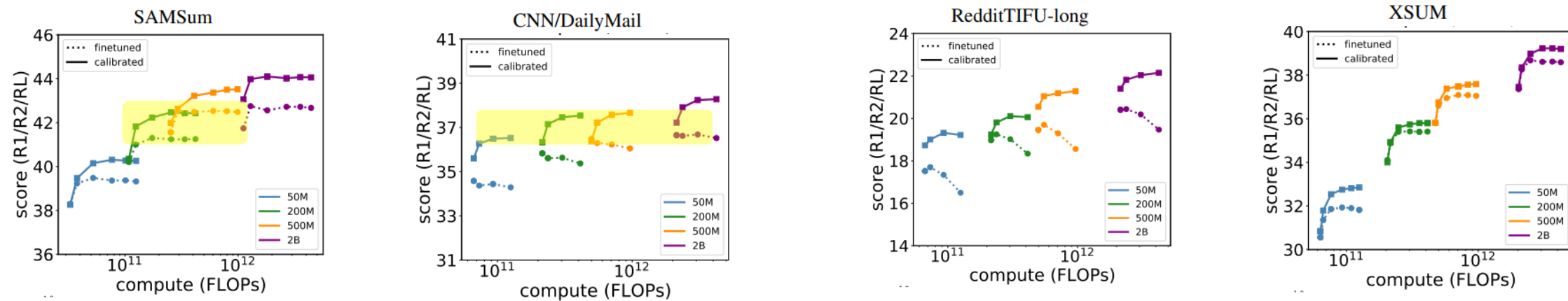


Figure 3. Quality and inference compute trade-off comparison between fine-tuned only and calibrated models.

Experiments

Final Result

4가지 language generation task에 대해 이전 SOTA와 성능 비교

Abstractive summarization, generative question answering, Question generation, data-to-text

	Dataset	#params	Prior SOTA	Our fine-tuned (2B)	Our calibrated (2B)
			R1 / R2 / RL	R1 / R2 / RL	R1 / R2 / RL
Abstractive summarization	CNN/DailyMail	340M ^a	47.78/23.55/44.57	44.31/21.91/41.41	47.97/24.18/44.88
	XSUM	268B ^c	–/27.1/–	49.57/26.77/41.41	49.77/27.09/42.08
	RedditTIFU-long	340M ^f	30.31/10.98/24.74 [*]	28.73/10.12/23.24	32.03/11.13/25.51
	SAMSum	20B ^b	–/29.60/–	53.64/29.21/44.83	54.37/29.88/45.89
Question generation	SQuAD QG	110M ^d	–/–/52.13	–/–/52.59	–/–/53.28
Question answering	MSMARCO NLG [†]	UNK ^e	–/–/69.77	–/–/70.73	–/–/71.06
data-to-text	WebNLG-en	20B ^b	–/55.40/–	76.96/52.97/62.56	78.09/55.52/65.06
	CommonGen [†]	20B ^b	–/37.40/–	66.49/36.17/58.82	68.95/38.49/60.13

Table 3. Calibrated PEGASUS_{2B} comparing with prior SOTA results

Contribution

1. SLiC를 제안함으로써 기존의 MLE Framework의 uncalibrated problem을 해결

Text generation task에서 SOTA 또는 그에 준한 성능을 보임

2. 새로운 calibration similarity metric 제안

BERTScore 수식 활용하여, context x 를 고려한 candidate, target 간 유사도 계산

3. Decoding heuristics에 대한 필요성 제거

Beam size optimization, Length normalization, Repetition prevention 등

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Calibrating Sequence Likelihood Improves Conditional Language Generation

Q & A

- Experiments
- Contribution 정리

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Calibrating Sequence Likelihood Improves Conditional Language Generation

Thank you